Problem set 1: Special relativity

1. **Some properties of vectors in Minkowski spacetime**
   Show that:
   (a) every vector that is orthogonal to a timelike vector is spacelike;
   (b) the sum of two future-directed timelike vectors is another future-directed timelike vector;
   (c) every spacelike vector can be written as the difference between two future-directed lightlike vectors.

2. **Relative Lorentz factor**
   Consider two observers O_1 and O_2 in uniform motion in an inertial frame, with respective 3-velocities $\vec{v}_1$ and $\vec{v}_2$ in that frame. Determine the Lorentz factor of O_2 in the rest frame of O_1.

3. **Some orders of magnitude**
   Using a Taylor expansion in terms of the speed β, prove that for small speeds, $\beta \ll 1$, the special Lorentz transformation corresponds to a Galilean transformation. In particular, prove that the law of additivity of velocities is recovered in this limit.
   Going to the next order in the Taylor expansion, find the first relativistic corrections to the Galilean transformation. Estimate the relativistic dilation of time relative to a clock at rest on the surface of the earth experienced by:
   - Usain Bolt;
   - a car;
   - an intercontinental flight;
   - a supersonic aircraft.

4. **Relativistic Doppler effect**
   Fill in the details of the derivation of the relativistic Doppler effect presented in the lecture notes, section 2.5.2.

5. **4-acceleration**
   Obtain Eqs (2.212)-(2.216) of the lecture notes, section 2.5.3.

6. **Variational principle and free motion**
   Fill in the details of the box "A variational principle for the free motion of particles" in section 2.5.3 of the lecture notes.

7. **Electromagnetic field**
   Recover Eqs (2.265)-(2.270) giving the transformation law for the electric and magnetic field under a Lorentz boost.

8. **Accelerated frames**
   Study in details section 2.7 of the lecture notes. In particular, rederive the results of 2.7.2 on Rindler observers.

9. **A beam of pions**
   Pions are subatomic particles which decay spontaneously with a half-life of 1.8×10^{-8} s. A beam of pions is accelerated until they move uniformly to a speed of $\beta = 0.99$ relative to the laboratory.
   (a) What time has elapsed in the laboratory before the intensity of the beam decreases by half?
   (b) Deduce the distance (in the laboratory) covered by the beam until its intensity has decreased by half.

   Note: The half-life corresponds to the time necessary for half of the particles at rest in the beam to have disappeared.

10. **Muon showers**
    Muons are leptons created in the high atmosphere, at an altitude $L \sim 15$ km, by the interaction of cosmic rays (protons) with the particles of the atmosphere. Their velocity acquired at creation, is almost equal to the ones of the cosmic rays, and of order $\beta = 0.99$. Muons are unstable particles which decay via weak interaction with a mean life $\tau \sim 2.19 \times 10^{-6}$ s such that the intensity of muons decreases with time as:
    $$I(t) = I_0 \exp\left(-\frac{t}{\tau}\right), \tag{10-1}$$
    where t is the time as measured in a reference frame at rest with the muons, and we have set $t = 0$ at the time of creation of the muons.
    (a) Justify that, in a Galilean invariant theory, most muons would have disappeared at sea level.
    (b) Making calculations in the reference frame of the muons, explain why special relativistic effects account for the possibility (at the actual fact) of observations of cosmic ray muons at sea level.
    (c) Give the 'symmetric' explanation in the reference frame of the earth.
    (d) Comment.